Md Sifat Hossain

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Research Interests

LLM-Based Code Generation and Automated Reasoning · Multi-Agent Systems for Autonomous Software Engineering · Cross-Language Program Repair and Polyglot Debugging · Benchmarking and Evaluation of Foundation Models · AI Alignment and Reinforcement Learning from Human Feedback (RLHF)

Education

University of Dhaka January 2020 – August 2026

Bachelor of Science in Computer Science and Engineering (CGPA: 3.13/4.00)

- **Thesis:** *A Hybrid LLM Feedback Framework for Automated Competitive Programming Workflows*
  Proposed a novel test-driven iterative benchmarking framework integrating multiple LLMs (OpenAI o3-mini, DeepSeek, Qwen) with Codeforces-based validation to evaluate and improve automated code generation and error correction in competitive programming contexts.
  Advisor: Md. Fahim Arefin, Assistant Professor, Dept. of CSE, University of Dhaka.
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Software Design Patterns, Artificial Intelligence, Probability & Statistics, Theory of Computation, Machine Learning, Natural Language Processing, Compiler Design, Operating Systems, Database Management Systems, Software Engineering

Publications

[J1] Anika Tabassum*, **Md Sifat Hossain***, Md. Fahim Arefin, Tariqul Islam, and Tarannum Shaila Zaman.
"A-ProS: Towards Reliable Autonomous Programming Through Multi-Model Feedback."
ACM Transactions on Software Engineering and Methodology (**TOSEM**), Vol. 35, No. 11, Article 357, pp. 1–40, November 2026.
`DOI: 10.1145/3820891` · `[PDF]` · `[Project Page]` · `[Code]`
*Equal contribution. Autonomous agentic framework separating solution generation (GPT-4/GPT-5) from specialized debugging feedback (DeepSeek-R1, Llama-3.3, Codestral) via persistent multi-model feedback loops. Evaluated on 367 ICPC and Codeforces problems; stateful refinement achieves 2.2–2.3× greater gains than stateless baselines and reduces error repetition by 2.9–3.5×.

[C1] **Md Sifat Hossain**, Anika Tabassum, Md. Fahim Arefin, and Tarannum Shaila Zaman.
"LLM-ProS: Analyzing Large Language Models' Performance in Competitive Problem Solving."
In *Proceedings of the 1st ACM International Workshop on Large Language Models for Code* (**LLM4Code**), co-located with ICSE 2025 – 47th International Conference on Software Engineering, Ottawa, Canada, pp. 80–87.
`DOI: 10.1109/LLM4Code66737.2025.00015` · `[PDF]` · `[Code]`
Novel evaluation framework benchmarking five state-of-the-art LLMs on 166 ICPC World Finals problems (2011–2024), revealing that o1 models significantly outperform general-purpose models due to chain-of-thought reasoning and iterative refinement capabilities.

[M1] **Md Sifat Hossain** et al. Manuscript on LLM agents for bugs that span two programming languages (Java+TypeScript, Python+Go). *Under double-anonymous review since October 2026; title withheld until January 2027.*

[M2] **Md Sifat Hossain** and Rick Kazman. "Do Explicit Architecture Decisions Help LLMs Keep Cross-Language Interface Contracts Consistent? A Registered Report." Stage 1 Registered Report, **MSR 2027** Registered Reports Track. *Manuscript submitted, November 2026.*

[M3] **Md Sifat Hossain** and Md. Fahim Arefin. "Code or Test? A Pre-registered Study of Which Artifact Language Models Side With When a Unit Test Fails." *Manuscript submitted, November 2026.*

[M4] **Md Sifat Hossain** et al. "CF-Live: What Gates Catch in a Post-Cutoff Competitive Programming Benchmark." *Manuscript submitted, November 2026.*

Research Experience

Remote Research Intern September 2024 – Present

RISE Lab (Reliable and Intelligent Software Engineering), UMBC *Remote*

- **Supervisor:** Tarannum Shaila Zaman (Assistant Professor, Dept. of Information Systems, UMBC).
- **Polyglot debugging:** developed LLM agents that find and fix bugs spanning Java+TypeScript and Python+Go code, built on a corpus of cross-language bug fixes mined from 100+ GitHub repositories and a SCIP-based cross-language navigation graph (∼90% precision). Paper under review ([M1]).
- Also contributed to **LLM-ProS** and **A-ProS**.

Research Assistant August 2023 – Present

Data Mining Research Lab, Dept. of CSE, University of Dhaka

- **Supervisor:** Md. Fahim Arefin (Assistant Professor, University of Dhaka).
- **LLM-ProS** (**ICSE 2025**, LLM4Code): built a benchmark of 166 ICPC World Finals problems (2011–2024) with automated Codeforces Gym submission, and compared five state-of-the-art LLMs on correctness, resource use, and reasoning.
- **A-ProS** (**ACM TOSEM 2026**): extended it into a multi-model agentic framework that separates solution generation from debugging feedback; built the orchestration pipeline, Codeforces browser automation, and verdict logging, and ran ablations on persistent vs. stateless context and critic calibration (ECE).
- **Thesis:** a test-driven pipeline pairing OpenAI o3-mini with specialist LLMs (DeepSeek, Qwen) to diagnose and fix code.

Competitive Coder Expert & Software Engineer June 2025 – Present

micro1 *Remote*

- Authored and reviewed 100+ competitive-programming tasks end-to-end (statements, editorials, reference solutions, test suites validated against brute-force oracles in sandboxed Docker verifiers), calibrated to defeat frontier models (avg@8 ≤ 0.4); also completed 100+ frontend evaluation tasks for a top AI lab at 94.8% accuracy.

- Authored 80+ containerized, terminal-based RL environments with hidden automated verifiers, including long-horizon cloud-infrastructure tasks where an agent deploys, repairs and tears down a multi-service system with Terraform/OpenTofu, scored on the live system; difficulty calibrated against frontier-model pass rates.

RLHF Data Researcher & Pod Lead July 2025 – Present
Turing Enterprises Inc. *Remote*

- Contributed to **Reinforcement Learning from Human Feedback (RLHF)** data creation for large-scale AI model alignment, focusing on dataset quality, consistency, and annotation methodology for code and reasoning tasks.
- **Benchmark project for a frontier AI lab:** authored and calibrated **49** containerized agentic-evaluation tasks, pairing realistic multi-file workspaces with deterministic verifiers (15–80 assertions each), tuned to a 25–75% pass band across four-rollout batteries.
- Fixed systemic defects in the team's validation tooling and packaging standards; wrote **9** standards documents and a ∼490-line Python validation gate that checked all 49 task packages for integrity defects before delivery.
- Led a team of 10 as pod lead, setting quality-control protocols and consistency standards that kept annotation reliable across task domains for downstream model training.

Research Software & Artifacts

LLM Benchmark Data Pipeline · `[Code]` Python, Selenium, BeautifulSoup, SQLite3

- Engineered the full data pipeline underlying the LLM-ProS and A-ProS publications: scraped 166 ICPC World Finals problems, normalized LaTeX/HTML content, and structured problem components into a consistent format for automated LLM prompt construction and verdict logging. Extended to 200+ Codeforces problems, forming the 367-problem A-ProS benchmark.
- Built SQLite-backed storage for per-attempt metadata (verdict, runtime, memory, iteration count), enabling reproducible experimental analysis across all model configurations.

TikTok Scraper · `[Code]` Python, Selenium, Requests, BeautifulSoup, SQLite3

- Engineered a scraper extracting video descriptions and author metadata for specified keywords and tags, enabling structured analysis across 5000+ videos. Same scraping architecture later reused in the LLM benchmark pipeline.

Industry Experience

Therap BD Ltd April 2025 – Present
Software Engineer *Dhaka, Bangladesh*

- Develop and maintain scalable features for Therap's EHR SaaS platform (used across 50 US states) using Java, Spring, Hibernate, JSP, and Oracle DB, with a focus on correctness and reliability under HIPAA compliance constraints.
- Build and containerize full-stack modules using ReactJS and Docker, deployed on WebLogic Server, contributing to platform stability and consistent delivery across multiple environments.

Honors & Awards

- **Zelf Hackathon 2.0** – Honorable Mention, Scraping Engineer track.
- **ICPC Dhaka Regional Onsite** – Top-35 of 220+ teams (2023); Top-49 of 309 teams (2024).
- **BUET Inter-University Programming Contest 2023** – 5th place among 102 university teams.
- **Samsung R&D BD Coding Contest 2024** – Final Round qualifier (rank 55 of 908 in Round 1).
- **Meta Hacker Cup 2024** – Round 2 qualifier (global rank 2166).
- **NCPC 2023** – 44th of 198 teams.
- **Competitive Programming** – Solved 1000+ problems across Codeforces (max. rating 1603, *Expert*), CodeChef (4★, max. rating 1921), AtCoder, and LeetCode.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, Kotlin
Frameworks & Libraries: Spring Boot, Hibernate, FastAPI, React.js, Next.js, JSP, Selenium, Playwright, Docker, Nginx
Databases: PostgreSQL, Oracle DB, MongoDB, SQLite
Research & ML Tools: PyTorch, HuggingFace Transformers, OpenAI API, Pandas, NumPy, Jupyter, LaTeX
Developer Tools: Git, Linux, Bash, Jenkins, JUnit5, WebLogic Server

Service & Leadership

- **Vice-President, ICT and Graphics**, Notre Dame English Club, Notre Dame College, Dhaka.
  Organized the 6th National English Carnival (10,000+ participants, 30+ events); previously co-ordinated the 5th edition (5,000+ participants). Led judge coordination, external communications, and sponsorship outreach.
- **Debater**, Gregorian Debating Club. Participated in school-level debate competitions.

Selected Software Projects

Poray Kemon · `[Code]` Next.js, TypeScript, PostgreSQL, Prisma, Redis

- Anonymous professor and course review platform for Bangladeshi universities: sign-in gates reviews, but the reviews table stores no user ID, so authorship cannot be joined back. 161-university catalog, moderation roles, automated tests.

Smart Event Ticketing System Java, Spring MVC, Hibernate/JPA, PostgreSQL, JSP

- Multi-role event management platform with pessimistic locking for transactional consistency; real-time event filtering with asynchronous data retrieval.

JobGenie · `[Code]` React.js, FastAPI, MongoDB

- Job search platform with automated CV generation and personalized job matching via live job scraping.

OyeAmigo · `[Code]` Kotlin, Android SDK

- Personality-based social networking Android app with null-safe Kotlin architecture.

References

Md. Fahim Arefin `fahim@cse.du.ac.bd`
Assistant Professor, Dept. of CSE, University of Dhaka
Supervised my thesis and both publications.

Tarannum Shaila Zaman `zamant@umbc.edu`
Assistant Professor, Dept. of Information Systems, UMBC
RISE Lab supervisor; collaborator on LLM-ProS and A-ProS.

Md Tariqul Islam `mtislam@umbc.edu`
Assistant Professor, Dept. of Information Systems, UMBC
Co-author of A-ProS.

Mia Mohammad Imran `imranm@mst.edu`
Assistant Professor, Dept. of CS, Missouri S&T
Research collaborator.